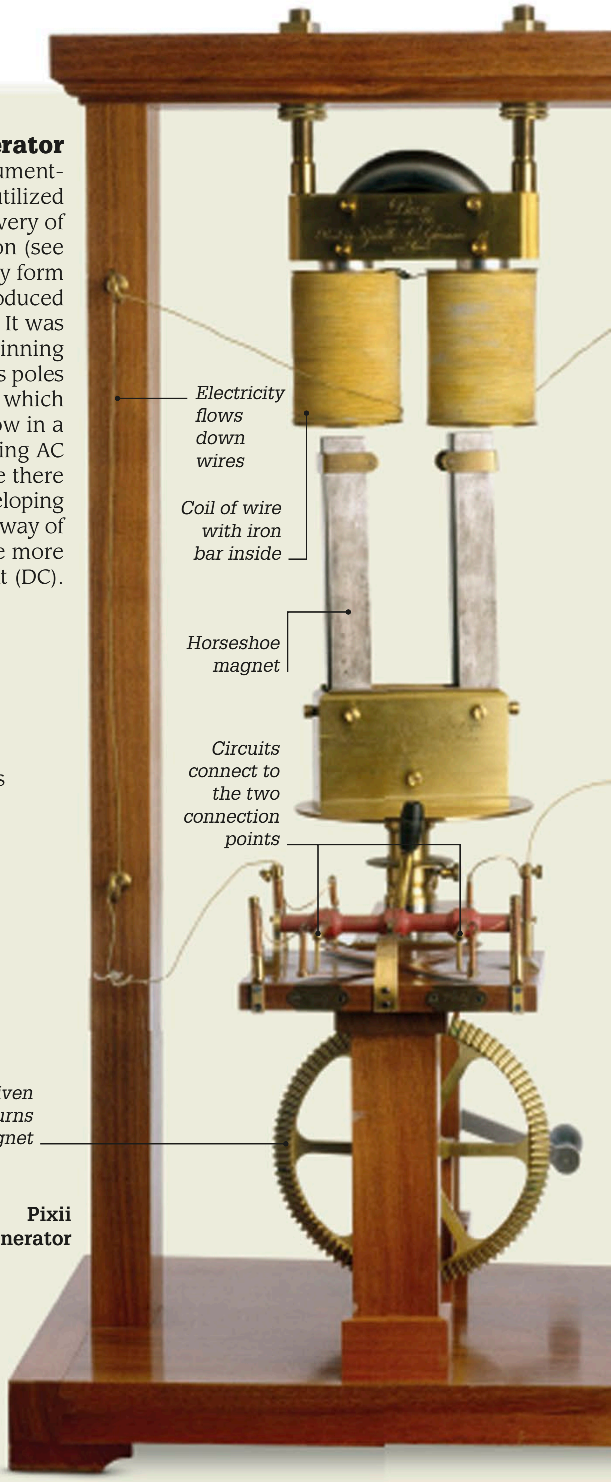
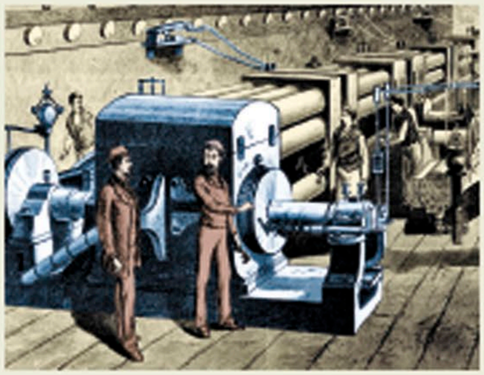
Powering our world

Today, electricity powers everything from huge industrial machines to our computers and smartphones, and lights our houses and cities. But people have only been using electricity as a source of energy for less than 150 years. Most of the electricity we use in our homes is made in power plants by huge turbines fueled by coal, gas, or renewable energy sources.

Early generator
In 1832, French instrument-maker Hippolyte Pixii utilized Michael Faraday's discovery of electromagnetic induction (see p.123) and built an early form of generator, which produced alternating current (AC). It was composed of a spinning horseshoe magnet, its poles pointing upward, which caused a current to flow in a coil of wire, generating AC electricity. At that time there was little interest in developing AC, and Pixii found a way of converting it to the more popular direct current (DC).



Pixii generator


The dynamo room at Pearl Street station

Lighting Manhattan
American inventor Thomas Edison's Electric Lighting Station, on Manhattan's Pearl Street, New York City, was the first permanent station to supply electric lighting on a grand scale. It featured six large dynamos that generated power to light more than 10,000 lamps. The dynamos were driven by steam engines and the steam was then used to heat nearby buildings. Pearl Street station burned down in 1890.

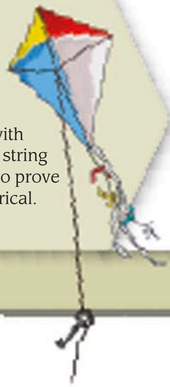
AC/DC: Different currents

Edison's power stations delivered direct current (DC) electricity. But DC could only travel a small distance before it decreased in power. In 1887, American engineer George Westinghouse introduced alternating current (AC), which could transmit electricity over longer distances.

- **Direct current**
  DC electricity flows in only one direction. It is used to charge batteries and as a power supply for electronic systems.
- **Alternate current**
  AC electricity changes direction many times every second. Most homes and businesses are wired for AC electricity.

Key events

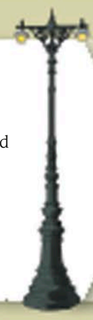
1752
Future American statesman Benjamin Franklin flew a kite with a key attached to the string into a thunderstorm to prove that lightning is electrical.



1800
Italian scientist Alessandro Volta made the first battery that could continuously provide electric current to a circuit. It was known as a Voltaic pile.

1831
English scientist Michael Faraday discovered electromagnetic induction when he found that he could produce an electric current by moving a magnet in and out of a coil of wire.

1882
Thomas Edison opened the first central power station for generating electricity on Pearl Street in Manhattan, New York City.


Electric street lamp